

AWS checklist and maintenance guide

Station name	M17	
People present	NCC MOLA	instructed by: ✓
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	31-7	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %: 5	

Notes:

flying helicopters / Tim

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	14 s $\pm$ 1
Adjust logger time to UTC (be sure PC runs on UTC time)	☒ Y / ☐ N
Download data to PC and/or change CF Card (wait for green light)	D/L + copy Y / ☒ N
Name of logger program	PROFICE 2012
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	N 65° 41.537
Longitude (dd mm.mmm)	W 39° 49.639
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / ☒ N
Mast tilt in boom direction (+ if radiometer tilting down)	+ 3.0 °
Mast tilt across boom looking from radiometer (- if clockwise)	+ 4.6 °
Boom direction relative to north	346 ° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	Y / ☒ N
Radiometer aligned with mast and inclinometer? If not, measure.	Y / ☒ N
Temperature / humidity height (bottom of casing – surface)	270 cm
Sonic ranger on AWS height (membrane – surface)	280 cm
Wind sensor height (center of propeller – surface)	320 cm
Sonic ranger on stakes height (membrane – surface)	157 cm
Free length of stake – outer, holding boom with sonic ranger	191 cm
Free length of stake – middle	199 cm
Free length of stake – outer, holding boom without sonic ranger	199 cm
Free length of ablation hose on/above the ice surface	10 m 53 cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	138 cm
Vertical difference in surface height (at station – at hose) (approx)	100 cm
Length of thermistor string on surface from surface marking	56 cm
Height of surface irregularities	0.4 cm
Stake assembly still standing?	☒ Y / ☐ N / more or less
Apparent damage to sensors? If yes, which?	Y / ☒ N

Notes:

Arrival 1235 LCC
Station looks good

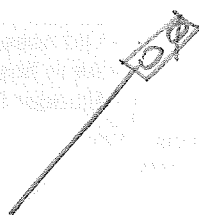
Depart. 15:46

Flesh card copied and reused, 2 GB,
data good (converted with CoriConverter)

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	☒ Y / N
Name of logger program	promice 2014d
Photo or screen dump of values in fast scan mode	☒ Y / N
Latitude (dd mm.mmm)	— JAT10
Longitude (dd mm.mmm)	— JAT10
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	2.3 °
Mast tilt across boom looking from radiometer (- if clockwise)	+ 4.7 °
Boom direction relative to north (radiometer should point to true south)	25 ° true/magn <i>see sketch</i>
Box on wind sensor exactly towards mast? If not, change.	☒ Y / N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	270 cm
Sonic ranger on AWS height (membrane – surface)	280 cm
Wind sensor height (center of propeller – surface)	320 cm
Sonic ranger on stakes height (membrane – surface)	87 cm
Free length of stake – outer, holding boom with sonic ranger	= cm
Free length of stake – middle	= cm
Free length of stake – outer, holding boom without sonic ranger	= cm
Free length of ablation hose on/above the ice surface	= cm
Alt: depth of ablation hose using markings on hose	= cm
Height of ablation hose fluid level above the ice surface	= cm
Vertical difference in surface height (at station – at hose) (approx)	= cm
Length of thermistor string on surface from surface marking	15 cm
Height of surface irregularities	40 cm

Notes:



Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	131307	=	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	415	=	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	GEUS 08	=	
Wind sensor	124021	=	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly			
T/hum assembly casing	TKRS: 6115493	=	- Fan spins and sounds OK: Y/N
T/hum assembly bracket	RS12T: 61148916	=	
Temperature probe in T/hum assembly	61154940	=	
HygroClip in T/hum assembly	61012684	61180279	- 40 °C offset in HygroClip temperature: Y/N <i>add one</i>
Sonic ranger on AWS	3885	=	- Old sensor with new membrane: Y/N <i>changed to</i> - Clicking 5 times as it should: Y/N <i>SAI CH</i>
Solar panel	C1070730946470	=	- Output OK: Y/N - Clean: Y/N
GPS antenna	018111213	=	- Stuck to top of logger enclosure: Y/N
Data logger	E3202	=	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	4590	=	- LED green of flashing orange (active): Y/N
Multiplexer	E4787	=	- Clicking as it should: Y/N
Iridium modem IMEI number	300039012259830	=	- Transmissions tested: Y/N - Problems: Y/N
Barometer	3766881		
Sonic ranger on stakes	5110 5110	=	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string	B16	B22	- Old string left in ice: Y/N/cut
Ablation hose assembly	5019ABS	=	- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		